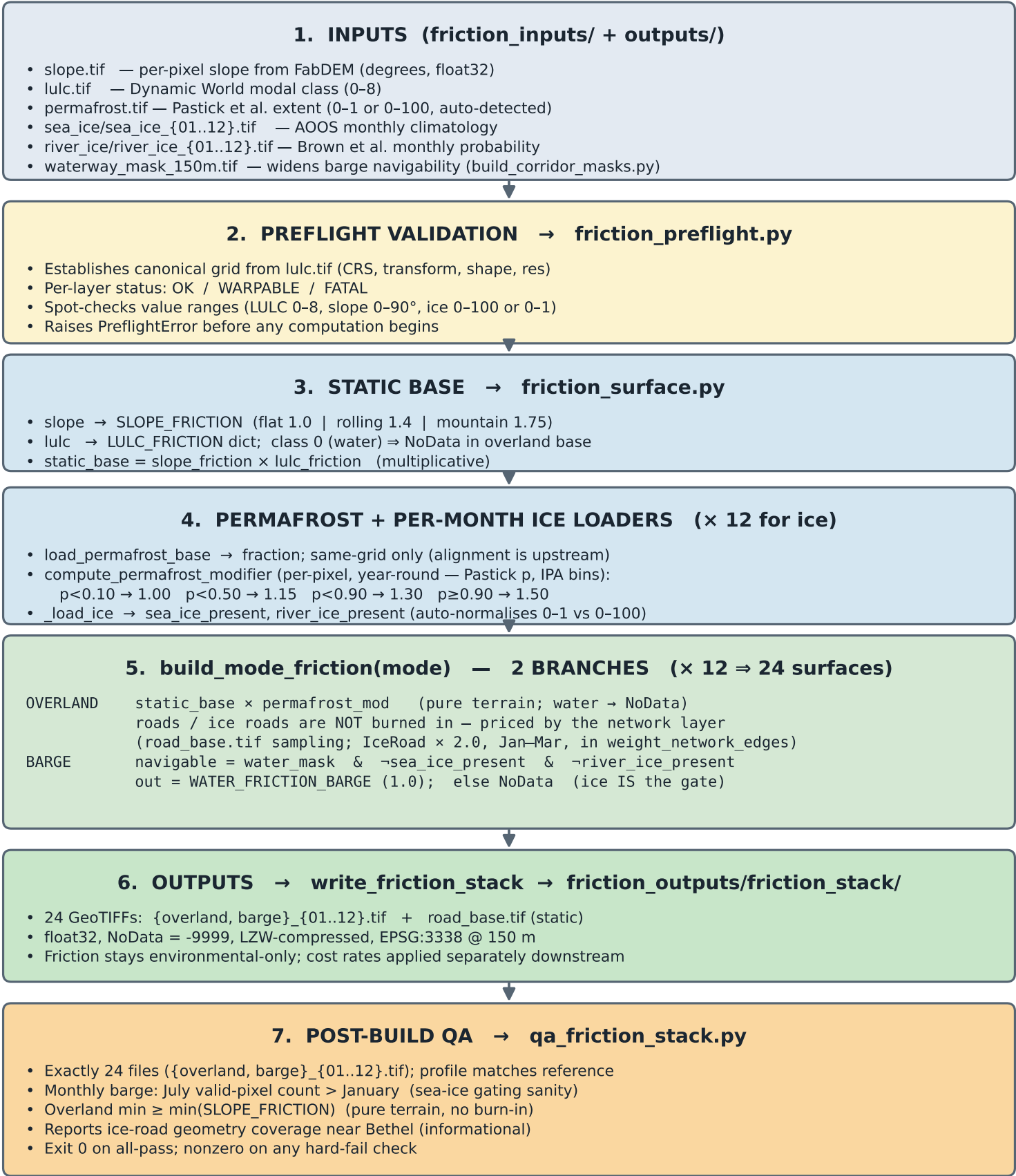


Friction Surface Construction Pipeline

friction_surface/ package · 2 modes × 12 months = 24 monthly raster surfaces



File-by-file responsibilities

friction_surface/ package — 11 modules

friction_config.py

Constants only — no I/O.

- CRS_TARGET = EPSG:3338, TARGET_RESOLUTION = 150 m
- SLOPE_FRICTION, LULC_FRICTION reclass tables
- PERMAFROST_ZONE_BREAKS = (0.10, 0.50, 0.90)
- PERMAFROST_ZONE_MULTIPLIERS = (1.00, 1.15, 1.30, 1.50)
- SEA/RIVER_ICE_THRESHOLD = 0.15 (NSIDC open-water limit)
- ROAD_FRICTION = 1.0 ROAD_BRIDGE_FRICTION = 1.0
- ICEROAD_TIME_PENALTY = 2.0 (UAF/INE 2023 Tbl 8.1)
- SEASON_MONTHS: ice road {1..3}, marine linehaul {6..10}
- MODES = ('overland', 'barge') (was 3, ice_road folded in)
- FRICTION_NODATA = -9999.0

friction_io.py

Generic raster I/O helpers shared by builders.

- load_and_ensure_crs(path, target_crs)
- load_raster_profile(path)
- Consumed by friction_surface, build_corridor_masks,
- build_connectors

friction_surface.py

The actual surface builder — pure numpy / rasterio.

- load_permafrost_base — same-grid permafrost loader
- compute_permafrost_modifier — Pastick p binned by IPA thresholds
- _load_ice — same-grid sea/river ice loader, auto-normalises units
- build_mode_friction(mode):
 - OVERLAND static_base x permafrost_mod (pure terrain)
 - BARGE navigable water; sea/river-ice gates
- compute_road_base → road_base.tif (network edge sampling)
- write_friction_stack(input_dir, output_dir) → 24 TIFs

friction_spatial.py

Nearest-feature geometry helpers (used by build_connectors).

- nearest_network_feature — point → nearest LineString
- nearest_point — point → nearest point-layer feature
- project_facilities_gdf — lon/lat → projected GeoDataFrame
- STATUS: orphan candidate — retires with
- build_connectors.py once the colleague's
- pre-snapped graph is adopted

qa_friction_stack.py

Post-build acceptance checks on the 24-raster stack.

- Exactly 24 files; profile matches reference
- Monthly barge: July > January valid pixels
- Overland: no negative values other than NoData
- Reports ice-road geometry near Bethel (info only)
- Exit 0 on all-pass; nonzero on hard-fail

viz/plot_friction_stack.py

Monthly-stack renderer — one 3x4 PNG per mode.

- Decimated read (~800 rows); shared 1-99 pct color scale
- Two-tone NoData underlay keyed on LULC water class:
 - water = ice blue, land = light gray
- Mode-specific legend spells out what NoData means —
 - barge: iced water above SEA/RIVER_ICE_THRESHOLD (impassable); overland: open water vs glaciers/DEM gaps
- Thresholds read live from friction_config (never hard-coded)
- Writes {mode}_monthly_friction.png to friction_outputs/

friction_paths.py

Single source of truth for filesystem layout.

- RASTER_DIR (env override)
- FRICTION_OUTPUT_DIR (env override)
- Raster paths: slope, lulc, permafrost, sea/river_ice
- Corridor masks: ICE_ROAD_MASK_TIF, ROAD_MASK_TIF,
- WATERWAY_MASK_TIF
- Networks: roads, waterways, airports, ports, flights
- Ice-road: ICE_ROADS_SHP, FUEL_DELIVERY_METHOD_*
- Chdirs to project root (opt-out: FRICTION_NO_CHDIR)

friction_preflight.py

Fail-fast validation of every raster before any work.

- Establishes canonical grid from lulc.tif
- LayerReport per file: OK / WARPABLE / FATAL
- Catches missing files, CRS / resolution / shape mismatches
- Sub-pixel offsets caught at ORIGIN_TOL_M = 0.01 m
- Value spot-checks: LULC ∈ 0..8, slope ∈ 0..90°,
 - ice ∈ 0..100
- Raises PreflightError on any FATAL

friction_costs.py

Cost rates & intermodal fees — env / econ split from friction.

- BASELINE_RATES_PER_GALLON_MILE (per mode)
- VEHICLE_MILE_RATES_REFERENCE (provenance)
- INTERMODAL_TRANSFER_FEES (per-handoff \$/gal)
- INTERMODAL_TRANSFER_ADDONS (blending, lightering)
- FUEL_DENSITY_LB_PER_GAL = 7.1
- STATE_CONTRACT_ADDERS_PER_GALLON (validation)
- MODE_METADATA, chain_cost_with_transfer_fees
- load_ice_road_communities(as_of) — historical-aware

run_friction_pipeline.py

End-to-end orchestrator.

- CLI: --input-dir, --output-dir, --skip-surfaces
- Validates inputs, calls write_friction_stack
- Edge step uses routing_wbt.compute_edge_costs
 - (legacy WBT cost-distance path)
- Writes mode_specific_edges to DuckDB +
 - connects_to backfill
- STATUS: WBT edge step retires under graph routing;
- surface-build step stays alive

test_friction_surface.py

Pytest suite covering the surface builder.

- load_permafrost_base: same-grid pass; raises on mismatch
- _load_ice: same-grid pass; raises on mismatch
- Ice threshold split invariant at equal thresholds
- Overland is pure terrain: static_base x permafrost
- Overland water pixels stay NoData
- Loader errors name the upstream align_* fix script